

Section 7.1 Worksheet

1. Verify the two eigenpairs for $\begin{bmatrix} 20 & 1 \\ 1 & 20 \end{bmatrix}$ using $(1, 1)$ and $(1, -1)$.
2. Find the characteristic polynomial and eigenvalues of $\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$.
3. Find a basis for each eigenspace in Problem 2.
4. For $B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, find the algebraic and geometric multiplicities of $\lambda = 2$.
5. Prove that E_λ is closed under scalar multiplication.
6. Find the eigenvalues of $\begin{bmatrix} 9 & 2 & -1 \\ 0 & 20 & 4 \\ 0 & 0 & 21 \end{bmatrix}$.
7. For $T(p) = p + xp'$ on P_2 , find $T(2 - 3x + 4x^2)$ and identify an eigenbasis.

Section 7.2 Worksheet

1. Prove that similar matrices have the same determinant.
2. For $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}$, verify the eigenpairs $(1, (1, 0, -1))$, $(2, (-1, 1, 1))$, and $(3, (-1, 1, 0))$.
3. Form P and D , and verify $P^{-1}AP = D$.
4. Use diagonalization to compute A^4 .
5. Determine whether $J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is diagonalizable.
6. Prove that eigenvectors associated with two distinct eigenvalues are linearly independent.
7. Explain why an $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Section 7.3 Worksheet

1. Determine whether $A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ is symmetric.
2. Prove that orthonormal columns imply $P^{-1} = P^T$.
3. Orthogonally diagonalize A from Problem 1.
4. Verify that the eigenvectors $(2, 1)$ and $(-1, 2)$ are orthogonal.
5. For $C = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, verify that $(1, 1, 1)$ is an eigenvector.
6. Apply one Gram-Schmidt step to $(1, -1, 0)$ and $(1, 0, -1)$.
7. Prove that every orthogonally diagonalizable real matrix is symmetric.

Section 7.4 Worksheet

1. Verify that $(20, 10, 5)^T$ is an eigenvector of $L = \begin{bmatrix} 0 & 6/5 & 2 \\ 11/20 & 0 & 0 \\ 0 & 11/20 & 0 \end{bmatrix}$ and find its eigenvalue.
2. Convert the stable age vector to proportions.
3. Solve $\mathbf{y}' = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \mathbf{y}$ with $\mathbf{y}(0) = (3, 1)^T$.
4. Find the symmetric matrix of $q(x, y) = 5x^2 + 4xy + 2y^2$.
5. Transform $5x^2 + 4xy + 2y^2 = 12$ to principal-axis coordinates.
6. Classify $q(x, y) = x^2 - 4y^2$.
7. Find the maximum and minimum of $5x^2 + 4xy + 2y^2$ subject to $x^2 + y^2 = 1$.
8. Prove that an orthogonal change of coordinates preserves length.